

Equity in skin typing: why it is time to replace the Fitzpatrick scale

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Dermatologists of colour have long championed skin of colour representation in education and workforce diversity.¹ For health equity, we must reconsider even fundamental and accepted terminology. The Fitzpatrick skin type (FST) scale, with both disproportionate focus on white skin tones and inconsistent use, perpetuates skin colour bias. Indeed, as FST does not purely objectively estimate skin pigmentation, it may inaccurately assess patients regarding risks for skin cancer and from interventions. Dermatology must seek an objective classification system, and given the rise of artificial intelligence (AI), technology-based approaches may be the solution.

The original Fitzpatrick scale was created in 1975 to classify reactivity of white skin to ultraviolet A phototherapy, although it remains unreliable for predicting minimal erythema dose. FST categorized skin colour into four types (I–IV) based on self-reported erythema and tanning reactions. The scale was later updated to include nonwhite patients, adding two categories initially based on colour alone and later with tanning reactions: ‘brown’ (type V) and ‘black’ (type VI). Now, as the standard skin classification system, FST is often incorrectly used as a proxy for race/ethnicity² and remains inherently biased as a scale centred on light skin.³

FST has been criticized for subjectivity, with several variables influencing results (e.g. word choices, assessment methods).⁴ Self-reported race and pigmentary phenotypes like FST also poorly predict sunburn risk.⁵ Further, FST does not accurately predict skin response to surgery or laser treatment, especially in darker skin, potentially leading to harmful outcomes (e.g. dyschromia) if relied upon.⁶

We must develop a system to stratify skin colour objectively rather than by tradition. Many established techniques allow physicians to describe skin quality and colour [e.g. Taylor Hyperpigmentation Scale and Mexameter® (Courage+Khazaka electronic GmbH, Cologne, Germany)]. However, these tools are either not accessible (e.g. Mexameter requires hardware), rely on human assessment, or are based on FST and may share limitations. We envision a system that, unlike FST, would not establish white skin as the baseline, and would identify subtle variations not visible to the naked eye.

AI may correct for bias in human assessments of skin colour. Algorithms could designate skin types based on an existing scale (e.g. Taylor Hyperpigmentation Scale), or AI can potentially create new scales by objectively assessing pigmentary scores without influence of perceived ethnicity, allowing more robust personalized clinical recommendations. Over a

decade ago, one of us (S.C.T.) convened a workgroup to create a classification system following this model. At the time, several barriers prevented development. Specifically, potential FST replacements lacked its simplicity. As such, any new system must be just as user-friendly and intuitive. Now, however, with new machine learning technology available, the potential for AI-based systems is evident.

The goal should be creation of a system representative of all skin types, yet just as important is making it logistically manageable in both clinical and research settings. To begin, we can leverage established frameworks, e.g. Taylor Hyperpigmentation Scale,⁷ a 150-hue gradation, as starting points for colour stratification. These scales combined with AI and objective pigment measures may allow rigorous and iterative algorithm development, establishing ground truth.

We envision a precise system classifying skin types from images or in-person via supervised machine learning. This system, developed with consensus-driven variables by expert panels, would expand current skin colour detection research and simplify programs into convenient instruments (e.g. smartphone applications) to guide physicians.

Still, barriers to replacing FST with AI models persist, notably machine learning's dependence upon human input, allowing for bias. Work investigating machine learning in dermatology has focused on melanoma screening using images of light skin, potentially leading to underperformance in skin of colour, and exacerbating disparities.⁸ Thus, any new system must involve a development process collaborative across disciplines, scrutinized for bias and over-indexed for images of darker skin tones. Further, any model's potential impact on clinical outcomes should be carefully studied before moving from proof-of-concept to bedside.

FST is so deeply rooted in dermatology that the idea of retiring it may seem disruptive, and some may question the bias and possible harm it perpetuates. However, this disruption is critical for meaningful change. Others might argue FST is useful in lighter skin and that new systems could jeopardize that utility. But should we be complacent with a flawed scale if better options are within reach?

As dermatology becomes more diverse, we must also move beyond the imprecise FST, a system rarely used for its original purpose and with debatable value. Rather than continuing a scale initially created for a specific group, we must commit to systems inclusive of all patients from inception. With AI, dermatology now has opportunities to lead efforts against bias in medicine, and replacing FST can be an important step.

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